Topic	Surface Water Modelling
Module	Hydrological Modelling
Assignment	Assessment of the effect of urbanisation on catchment hydrology and identification of suitable mitigation measures

Due 12:00 pm on 27 March 2026 (online submission on LEARN: report in .pdf format with the filename convention Lastname_Firstname_HMAr_ENCN442.pdf)
The assignment is worth 10% of the final grade.

Project description

You are a water resources engineer working for an engineering consulting company. The city council plans the development of a new subdivision in an existing township. Your company has been hired to provide an assessment of the effects of the subdivision development on catchment hydrology. You have been provided with catchment characteristics, information on potential measures that the city council would like to see investigated and related data.

Tasks

Provide an assessment of the effect of urbanisation on catchment hydrology. Evaluate potential measures to mitigate negative effects. The goal is to reduce the discharge at the catchment outlet to at least pre-development level.

Develop a rainfall-runoff model for the catchment using HEC-HMS in order to analyse and assess the effects of urbanisation. The catchment is divided into four sub-catchments (see Figure 1). Sub-catchment 4 is undergoing urbanisation due to the development of a new residential area. Potential measures that the council would like to see investigated are downstream channel modifications. The intended purpose of these measures is to reduce the effects of the additional flow resulting from the development.

Determine, assess and compare the runoff hydrograph at sensible locations in the catchment and at the catchment outlet for these different conditions:

(1) the existing conditions,
(2) the existing condition in sub-catchments 1 to 3 with sub-catchment 4 urbanised,
(3) the same as (2) but investigating a modified channel in sub-catchment 1.

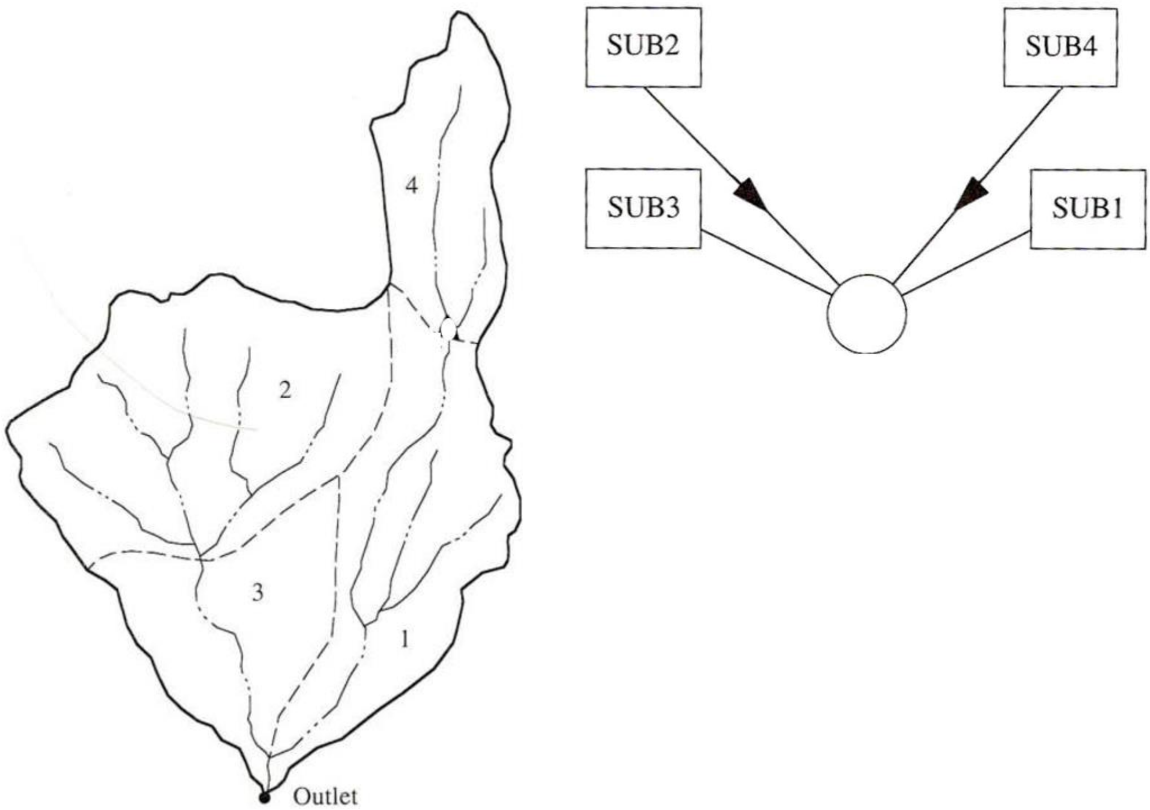

Figure 1: Study catchment, subdivided into four sub-catchments (SUB1, SUB2, SUB3 and SUB4). The catchment outlet is also shown. On the left is the GIS delineation of the catchment boundary, the sub-catchments and the streams. On the right is a schematic representation of the catchment.

Perform subarea runoff computations by applying Snyder's synthetic unit hydrograph with rainfall loss rates determined using the SCS curve number method, and carry out channel routing with the Muskingum method. Table 1 lists the current characteristics of the sub-catchments.

Table 1: Characteristics of the four sub-catchments.

Sub-catchment	Area [km^2]	Catchment length [km]	Length to centroid [km]	SCS curve number, CN
1	3.94	4.26	2.25	70
2	5.62	2.98	1.09	84
3	2.48	1.82	0.97	80
4	2.23	2.40	1.27	70

The parameters for Snyder's synthetic unit hydrograph for the existing condition are peaking coefficient C_p = 0.25 and lag coefficient C_t = 0.38. The flood wave travel time (Muskingum coefficient K) for the stream reach passing through subarea 3 is estimated as 0.3 h, and the travel time for subarea 1 is estimated as 0.6 h. The Muskingum *X* has been approximated as 0.2 for each of the two stream reaches.

The design rainfall is a hypothetical 0.1 AEP storm defined by the depth-duration data listed in Table 2. Note: use the method 'Frequency Storm' in HEC-HMS. The interpolation equations for the 10- and 30-minute durations are

$$P_{10-min} = 0.59P_{15-min} + 0.41P_{5-min}$$
$$P_{30-min} = 0.49P_{60-min} + 0.51P_{15-min}.$$

Table 2: Design rainfall depth-duration data.

Duration	5 min	15 min	1 h	2 h	3 h	6 h	12 h	24 h
Rainfall (mm)	9.7	18.8	33.0	43.2	53.3	76.2	127.0	177.8

A residential development in sub-catchment 4 will increase the impervious area so that the developed SCS curve number will be 85. The unit hydrograph parameters are expected to change to C_t = 0.19 and C_p = 0.5. Modification of the channel through sub-catchment 1 will change its Muskingum routing parameters to *K* = 0.4 h and *X* = 0.3.

Structure of the report

Note that the report should be written in 'memo-style'. This is commonly done to summarise results for internal company use (documentation of progress, update for the line manager or CEO, etc.), i.e. this is not a full client report. The memo might later become part of the larger client report. So, while this is an interim, brief document, it must still be written to the highest standard, with high quality figures and tables.

The report must be structured in the following manner:

1. Executive summary and key recommendations (max. ½ page)
2. Modelling approach and assumptions
3. Pre-development conditions
4. Post-development urbanised conditions
5. Channel modification

Marking scheme (100 marks total)

- Executive summary and key recommendations: 20 marks
- Current situation and assessment of the effect of urbanisation on catchment hydrology: 40 marks
- Evaluation of potential measures as suggested by the city council to mitigate negative effects: 30 marks
- Report presentation: 10 marks

General remarks

- Clearly state and justify any assumption you make.
- The report must have a title, date and the name and email address of the author.
- Use adequate labels for the figures, thereby including physical units where applicable. Axes must be labeled with the name of the variable and the unit (if any). If there is more than one set of points or curves on a Figure, a legend must be added showing clearly what each one is.
- All figures must be scaled so the points, curves, etc. can be seen properly.
- Make sure that for your figures the font size has been chosen such that all numbers, labels, symbols, etc. are clearly legible.
- All figures and tables must have a numbered caption. Refer to figures and tables in the text as e.g. 'Figure 1 shows…'…'listed in Table 1'…
- Note that a quantitative answer is more desirable than a qualitative answer. Hint: When comparing solutions, provide e.g. a percent reduction/increase in discharge and use various types of figures and tables to present your results).
- **Page limit: 5 pages (title page does not count towards the page limit; do not include a table of contents).** Aim at keeping the report brief and to the point. You will write a great many of such reports in your future career and it is important that you practice conveying the key messages in a concise manner. Do not show all results, but those that led to the best outcome. Produce meaningful and information-rich figures and tables. Despite being concise, make sure that the reader can fully understand what has been done to arrive at a particular result.
- Page Margins: Minimum 20 mm
- Font and Font Size for the main body of the report: Calibri, 12 point
- Font and Font Size for Figure and Table captions: Calibri, 11 point
- Number the pages.
- Marks will be deducted for exceeding the page limit, figures that are not legible, incorrect page margins, incorrect font size, incorrect file names, incorrect file format, not submitting all files requested.
- Lastly: There is a penalty for late submission.

Further information

A short video on how to use Snyder's UH can be accessed here, provided by the Hydrologic Engineering Centre of the USACE: https://www.youtube.com/watch?v=ZuSj53-H8eY

The HEC-HMS documentation is an excellent resource:
https://www.hec.usace.army.mil/software/hec-hms/documentation.aspx